Large Data Set: OCR MEI (World Bank)

Questions 1–13

Question 1. These exercises were written using LDS7 but should continue to be of use for future World Bank Large Data Set samples. State the following facts about the Large Data Set:

a) The country is the official name used by which country?
b) Which regions are included in the data?
c) How is the population of each country found?
d) What is the unit of the total area of a country?

Question 2. Explain what a birth or death rate per 1000 is and what point of the year it is based on.

Question 3.

a) Explain what GDP is.

b) What does 'real GDP per capita' mean?

Question 4. Explain what is meant by 'physician density' and what definition a physician has for this data set.

Question 5.

a) The World Health Organisation categorises healthcare workers as what?

b) According to the World Health Organisation, fewer than how many healthcare workers per 1000 would be insufficient to achieve coverage of primary healthcare needs?

Using your answer to part b)

c) Identify one of the 3 countries in Europe that do not meet the World Health Organisation value for coverage of primary healthcare needs.

d) Identify the region of the world in which no countries meet the World Health Organisation value for coverage of primary healthcare needs.

Question 6. Explain what is meant by 'current health expenditure' and why it is a useful piece of data.

Question 7. Caspian claims that the world has more mobile phones than people.

a) Explain how the Large Data Set indicate that this may be the case in some countries.

b) In which region are there the fewest mobile phone subscribers per 100 people in the population?

Caspian would like a sample of 46 countries to test his theory.

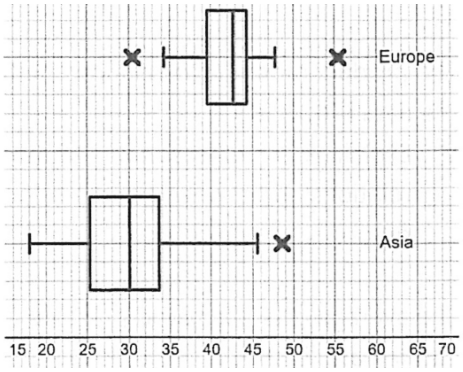
c) Explain how he could take a sample which proportionally represents each of the regions in the world.

Question 8. Jamie is comparing the life expectancy in 2020 to the life expectancy in other years. Are there any countries in which life expectancy went down in the time frame from

a) 1960 to 2020?

b) 1990 to 2020?

Question 9. The median age of countries in Europe and Asia are shown in the boxplots below:



a) Identify the outliers in each of the boxplots.
b) The Europe data had to be cleaned before it could be used, explain why this is the case and how it could be identified in the Large Data Set.
c) Make a comparison of the two data sets.

Oscar states that there is a country missing from the boxplots.

d) Which country was not included in either boxplot, but lies in the correct part of the world?

Question 10. The table below shows the current health expenditure for the Caribbean after it has been cleaned:

Country	CHE (%)
Antigua and Barbuda	4.4
The Bahamas	5.8
Barbados	6.3
Cuba	11.3
Dominica	5.5
Dominican Republic	5.9
Grenada	5
Haiti	4.7
Jamaica	6.1
Saint Kitts and Nevis	5.4
Saint Lucia	4.3
Saint Vincent and the Grenadines	4.8
Trinidad and Tobago	7

a) Find the mean and the standard deviation of the current health expenditure for the Caribbean.

Any value more than 2 standard deviations from the mean can be identified as an outlier.

b) Identify any countries which are outliers.

The current health expenditure in Oceania is

$$\bar{x} = 8.79\% \text{ and } s_x = 5.87\%$$

c) Compare these to the statistics calculated for the Caribbean above.
d) Give two reasons why it may be fair to compare the Caribbean with Oceania.
e) Give a reason why it may not be fair to compare the Caribbean with Oceania.

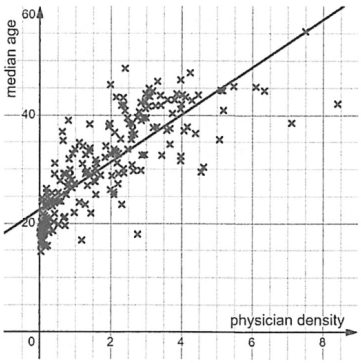
Question 11. The life expectancy at birth, L, between 1960 and 2020 can be modelled using a straight line where t is the number of decades since 1960.

$$\begin{aligned} \text{Botswana:} \quad & L = 2.01t + 50.1 \\ \text{Chile:} \quad & L = 3.90t + 55.5 \end{aligned}$$

a) Predict the life expectancy for 1995 for both countries.
b) Explain what each of the values in the models mean for each country.
c) Compare the change in life expectancy of the two countries.

Question 12. The median age of a country is plotted against the physician density (physicians/1000 population) of a country, creating the graph below. The product moment correlation coefficient is $r = 0.7949$ and the least squares regression line is given by

$$y = 4.37x + 22.6$$



a) Comment on the value of r and what it means for the association between median age and physician density.

It is known that Cuba has a very high physician density but not a high median age.

b) Identify which is likely to be the point which represents Cuba.

A country which was not included in the graph decides to manage their physician density, so it is 7 per 1000 people with the aim to increase the median age of people in their country.

c) Use the regression line to determine the median age for this country.
d) Comment on your result in part c) with reference to the aim of the country.

Question 13. Prerequisite knowledge for Q13: Normal Distribution The death rate per 1000 people across the world may be considered normally distributed, where $X \sim \mathrm{N}(7.5, 6.92)$. Find

a) the probability that the death rate per 1000 people of a country is higher than 6
b) $\mathrm{P}(\mu - 2\sigma < X < \mu + 2\sigma)$
c) the death rate per 1000 for the 10% of the countries with the highest death rate.